

Attendance Risk Calculator — main.py

Complete Python source code for the Attendance Risk Calculator project.

```
def calculate_attendance(total_classes, attended_classes):
    """Calculate attendance percentage and risk level."""
    if total_classes <= 0:
        raise ValueError("Total classes must be greater than zero.")

    if attended_classes < 0:
        raise ValueError("Attended classes cannot be negative.")

    if attended_classes > total_classes:
        raise ValueError(
            "Attended classes cannot be greater than total classes."
        )

    percentage = (attended_classes / total_classes) * 100

    if percentage >= 75:
        risk = "Low Risk"
        recommendation = "Attendance is satisfactory."
    elif percentage >= 60:
        risk = "Medium Risk"
        recommendation = "Attendance needs improvement."
    else:
        risk = "High Risk"
        recommendation = "Immediate attendance improvement is recommended."

    return percentage, risk, recommendation


def main():
    print("=== Attendance Risk Calculator ===")

    try:
        total_classes = int(input("Enter total classes conducted: "))
        attended_classes = int(input("Enter classes attended: "))

        percentage, risk, recommendation = calculate_attendance(
            total_classes, attended_classes
        )

        print(f"\nAttendance Percentage: {percentage:.2f}%")
        print(f"Risk Level: {risk}")
        print(f"Recommendation: {recommendation}")

    except ValueError as error:
        print(f"\nError: {error}")


if __name__ == "__main__":
    main()
```